

BNOCR

BITCOIN NODE OPERATIONAL CONTINUITY RESEARCH

Technical Research Concept

Proposed by: UBRO CLASSIFIEDS LIMITED

Programme: BNSAT-R - Bitcoin Node Security, Availability & Resilience
Research

Location: Kampala, Uganda

Research. Measure. Build. Contribute.

1. Research Concept

BNOCR proposes to investigate the operational recovery and continuity of independently operated Bitcoin full nodes following controlled disruptions.

The project is based on the observation that a Bitcoin full node does not simply transition between two states of working and not working. Following a disruption, several recovery processes may occur, including system recovery, process recovery, network recovery, peer discovery, blockchain synchronisation and return to normal operation.

BNOCR proposes that these stages may be measurable and that understanding them may provide useful information about Bitcoin infrastructure operational resilience.

2. Primary Research Question

How do independently operated Bitcoin full nodes recover and return to operational continuity following controlled disruptions, and can this recovery process be measured using a reproducible framework?

3. Research Hypothesis

The project will investigate the hypothesis that:

Bitcoin full-node recovery behaviour varies according to the type and severity of operational disruption, node resource profile and surrounding infrastructure conditions, and that these differences can be measured using a reproducible recovery-state framework.

This is a hypothesis to be tested. The research will not assume that the hypothesis is correct.

4. Proposed Recovery State Model

R0 - Failure: Node unavailable.

R1 - Process Recovery: Bitcoin Core process successfully starts.

R2 - System Recovery: Underlying operating environment is functioning normally. R3

- Peer Recovery: Normal peer connectivity is restored.

R4 - Chain-State Recovery: Node returns to the expected blockchain state. R5

- Operational Recovery: Node resumes normal operation.

R6 - Verified Continuity: Node remains stable for a defined observation period.

5. Experimental Conditions

The research may investigate controlled conditions involving:

Power: Graceful shutdown; unexpected power loss; repeated interruption.

Network: Complete connectivity loss; temporary connectivity loss; intermittent connectivity.

Bandwidth: Reduced bandwidth; variable bandwidth; high-latency conditions.

Computing Resources: CPU constraints; memory constraints.

Storage: Storage pressure; low available disk space.

The final experiments will be selected after the prior-work review.

6. Experimental Design

The project will use controlled laboratory environments.

Each experiment will record node configuration, hardware profile, software version, baseline state, disruption type, disruption duration, recovery timestamps, resource usage, errors, manual interventions and final recovery state.

Experiments will be repeated to assess consistency.

7. Research Metrics

Time-Based Metrics: Time to process recovery; time to peer recovery; time to chain-state recovery; time to operational recovery.

Reliability Metrics: Recovery success; recovery failure; recovery consistency.

Resource Metrics: CPU; memory; storage; network bandwidth.

Human Factors: Manual intervention; number of intervention steps; recovery complexity.

8. Proposed Research Output

The central output will be a Bitcoin Node Operational Continuity Framework.

The framework may include recovery states, definitions, metrics, experimental methodology, data format and benchmark procedures.

The objective is to create a framework that can potentially be used by independent researchers and node operators.

9. Potential Open-Source Tools

Subject to research findings, UBRO may develop a Node Recovery Monitor, Recovery Benchmark Tool, Experimental Automation, Data Analysis Tools and a Recovery Dashboard.

UBRO will first investigate whether equivalent tools already exist. Where existing tools are suitable, UBRO will seek to use or contribute to them rather than duplicating them.

10. Potential Research Contribution

The project could contribute to Bitcoin infrastructure research, node operational knowledge, open-source tooling, disaster recovery practices and infrastructure resilience research.

The research may also provide practical information for operators in environments where infrastructure reliability is variable.

11. Research Limitations

Laboratory experiments cannot fully represent global Bitcoin infrastructure.

Results from Uganda should not automatically be generalised globally.

Hardware profiles may not represent all node operators.

Experimental conditions may simplify real-world failures.

The research will therefore clearly distinguish observed results, experimental findings, hypotheses and generalisations.

12. Future Research

Depending on findings, future research could investigate multi-site node recovery, mobile connectivity, low-cost hardware, solar-powered node deployments, remote recovery, disaster recovery, offline operational procedures and infrastructure redundancy.

These areas will only be pursued if supported by research evidence.

13. Research Position

BNOCR does not propose to modify Bitcoin's consensus rules.

It does not propose changes to Bitcoin governance.

It does not seek to centralise Bitcoin infrastructure.

It seeks to provide independent evidence concerning the operational behaviour of Bitcoin full nodes.

The research is therefore positioned as an open-source infrastructure and security research contribution.

14. Conclusion

BNOCR proposes a focused investigation into the operational continuity and recovery behaviour of Bitcoin full nodes.

The research begins with a question rather than a predetermined conclusion.

Its success will be measured by the quality of its methodology, the reproducibility of its experiments, the usefulness of its findings and the quality of its contribution to the open-source ecosystem.

UBRO intends to seek technical feedback from experienced Bitcoin developers and researchers before finalising the experimental methodology.

Status and Request for Ecosystem Feedback

Status: Research Concept - Seeking Technical Review and Ecosystem Feedback

This document represents an initial research concept and is not presented as a claim of established novelty. UBRO CLASSIFIEDS LIMITED welcomes feedback from Bitcoin developers, researchers and open-source contributors to refine the research direction and avoid duplication of existing work.

UBRO is particularly interested in technical guidance concerning whether the proposed research question is meaningful and sufficiently distinct, whether the proposed recovery-state approach is useful, what existing projects or literature should be reviewed, and how the research can best contribute to the wider Bitcoin ecosystem.